

Governing Eqs - Applied: Boussinesq, Incompressibility and the Speed of Sound, Vortex Stretching, Biot-Savart Law, Rankine Vortex

→ Mathematical expressions of the three fundamental conservation laws that govern fluid flow: **mass, momentum, and energy**.

- **Navier-Stokes:** Includes viscosity, complex.
- **Euler:** No viscosity, simplified.
- **Bernoulli:** Steady-state, energy-based, derived from Euler.

AA 507 (Incompressible Fluid Mechanics) Winter 2026, UW

Assignment: Homework 4

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clear % Clears workspace

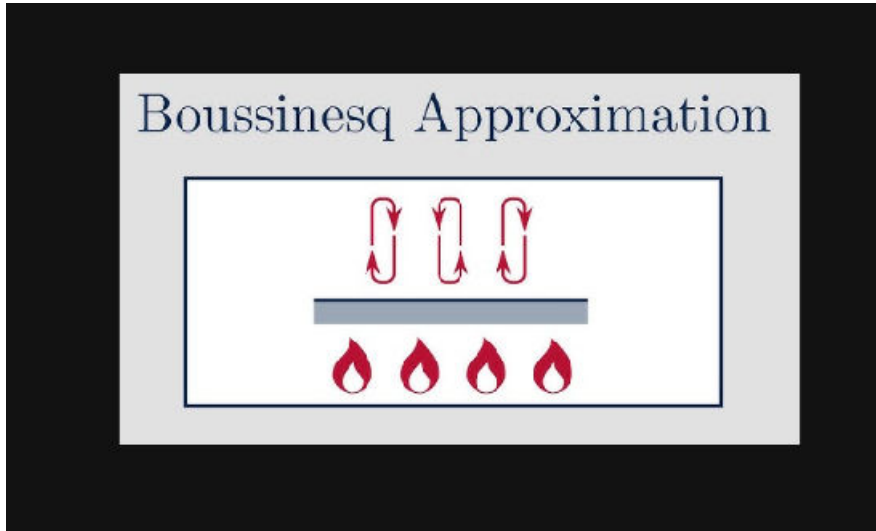
```
clc % Clears command window
close all % Closes all figures
```

Problem 1. Boussinesq Approximation and Dimensional Analysis: Designing a Lava Lamp

Background

The Boussinesq approximation is a simplification to model **buoyancy-driven** flows (natural convection),

```
imshow("Boussinesq_approx.png")
```

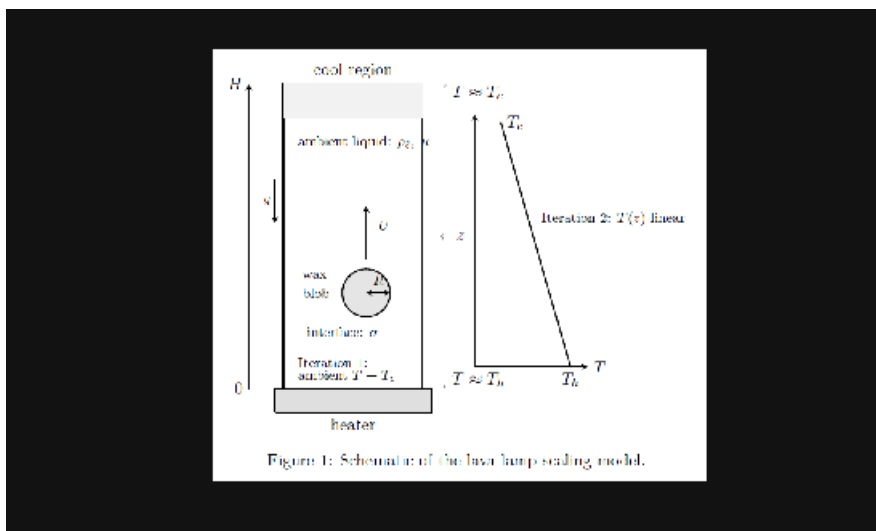


which in modeling fluid flow - such as CFD, this becomes much less computationally expensive than solving the full compressible formulation of the Navier-Stokes equations.

Problem statement:

In the problem given, a **lava lamp** scaling model is given:

```
imshow('lava_lamp_scaling_model.png')
```



Big Idea:

The Boussinesq approximation simplifies **buoyancy-driven flow equations** by treating fluid density as constant (incompressible) everywhere except in the gravity-driven buoyancy term.

Applications:

The assumption of small temperature/composition variations, and treating density changes as linear with temperature, makes it ideal for modeling natural convection, atmospheric, and ocean flows.

Key Assumptions

- Small density variation -- change in density $\Delta\rho$ is very small compared to the reference density ρ_0 .
- Incompressible continuity -- density changes are small, the fluid is treated as incompressible.
- Linear equation of state -- density is assumed to vary linearly with temperature, e.g., the density of the wax:

$$\rho_w(T) \approx \rho_{w,c}[1 - \beta(T - T_c)] \quad (1-1)$$

where β is the thermal expansion coefficient.

Physical significance: The Boussinesq approximation assumes the fluid's **inertia** is constant, but its **weight** (gravity term) changes enough to drive motion.

To that end, the buoyancy-producing density difference scales as:

$$\Delta\rho \sim \rho_{w,c}\beta\Delta T_{drive} \quad (1-2)$$

- [Means: $\Delta\rho$ "scales with" ...]

where ΔT_{drive} is the temperature mismatch between the wax blob and the surrounding/ambient fluid, the "driving temperature."

Iteration 1 (Two-Temperature Model)

The ambient fluid/liquid is everywhere T_c and the wax blob starts at T_h , so the temperature mismatch is:

$$\Delta T_{drive} \sim \Delta T \equiv T_h - T_c \quad (1-3)$$

- [The driving temperature "scales with" the change in temperature, which **"identically equal to"** ($\equiv$) the temperature difference between T_h and T_c ,
- identically equal to means the equality **holds for all values** of the variables, not just specific cases.]

Analysis

1. Estimating the wax blob radius R

The bouyant force is given [1]: $F_B = \rho_f g V$

The bouyant force **scale** of the wax blob, then becomes:

$$F_b \sim \Delta\rho gV$$

Where $V \sim R^3$, ignoring numerical prefactors.

$$\rightarrow F_b \sim \Delta\rho gR^3 \quad (1-4)$$

The surface tension force scale:

$$F_\sigma \sim \sigma_s R$$

Force balance: $F_b \sim F_\sigma$

$$\rightarrow \Delta\rho gR^3 \sim \sigma_s R \quad \rightarrow \quad R = \left(\frac{\sigma_s}{\rho_{w,c}\beta\Delta T_{drive}g} \right)^{1/2} \quad (1-5)$$

```
%% HW4 Problem 1 (Iteration 1, Part 1)
% Plot blob radius R vs surface tension sigma using Symbolic Math Toolbox
% Scaling: R = sqrt( sigma / (rho_wc * beta * DeltaT * g) )

%clear; clc; close all;

% --- Symbolic setup ---
syms sigma rho_wc beta DeltaT g positive
R_sym = sqrt( sigma / (rho_wc*beta*DeltaT*g) ); % symbolic expression

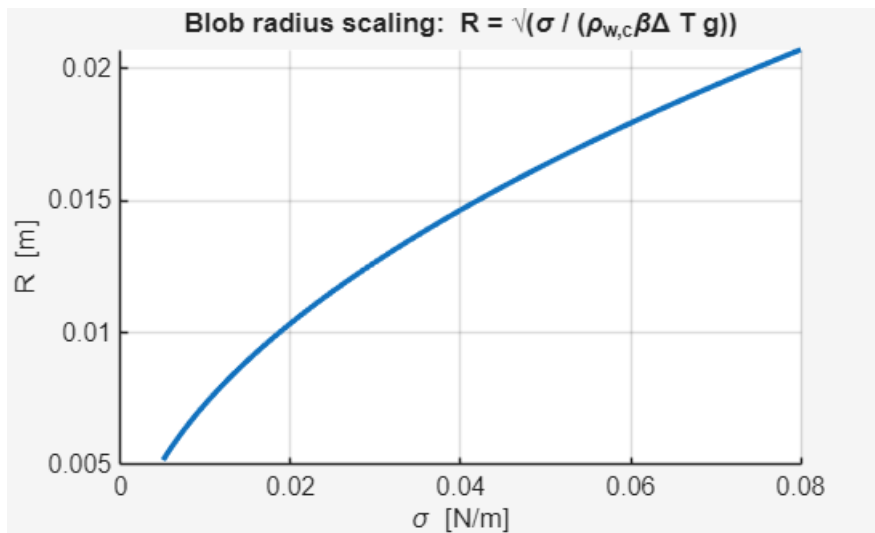
% --- Numerical parameters (edit as needed) ---
rho_wc_val = 950; % [kg/m^3] (example wax density near Tc)
beta_val = 1e-3; % [1/K] (example thermal expansion coefficient)
DeltaT_val = 20; % [K] (Th - Tc)
g_val = 9.81; % [m/s^2]

% --- Sigma sweep (example range) ---
sigma_min = 0.005; % [N/m]
sigma_max = 0.080; % [N/m]
N = 300;
sigma_vec = linspace(sigma_min, sigma_max, N); % [N/m]

% --- Evaluate symbolically -> numeric ---
R_fun = matlabFunction(R_sym, "Vars", [sigma, rho_wc, beta, DeltaT, g]);
R_vec = R_fun(sigma_vec, rho_wc_val, beta_val, DeltaT_val, g_val); % [m]

% --- Plot ---
fig = figure; hold on;
fig.Theme = "light";
plot(sigma_vec, R_vec, 'LineWidth', 2);
grid on;
xlabel('\sigma [N/m]');
ylabel('R [m]');
```

```
title('Blob radius scaling: R = \surd(\sigma / (\rho_{w,c}\beta\Delta T g))');
```



```
% --- (Optional) display symbolic formula in Command Window ---
disp("Symbolic formula for R(sigma):");
```

Symbolic formula for R(sigma):

```
pretty(R_sym)
```

$$\frac{\sqrt{\sigma}}{\sqrt{\Delta T} \sqrt{\beta} \sqrt{g} \sqrt{\rho_{wc}}}$$

Physical Interpretation

As the radius of the wax blob increases, so does the surface tension, which makes for a more **stable blob**.

2. Estimating the thermal equilibrium time scale

2.1 First Principles: Start from the Heat Equation (Fourier)

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$$

Where,

α is the thermal diffusivity

∇^2 is the Laplacian

T varies over the blob of characteristic size R

2.2 Scaling Analysis

Estimate orders of magnitude

Time derivative scale: $\frac{\partial T}{\partial t} \sim \frac{\Delta T}{t_{th}}$

since the temperature changes by ΔT over the characteristic time t_{th} .

Laplacian scale: Temperature varies over the length scale R --

$$\frac{\partial T}{\partial x} \sim \frac{\Delta T}{R}$$

Second derivative: $\frac{\partial^2 T}{\partial x^2} \sim \frac{\Delta T}{R^2}$

Since the Laplacian is the sum of second spatial derivatives,

$$\nabla^2 T \sim \frac{\Delta T}{R^2}$$

2.3 Substitute the scalings into the heat equation

$$\frac{\Delta T}{t_{th}} \sim \alpha \frac{\Delta T}{R^2}$$

Cancel ΔT :

$$\frac{1}{t_{th}} \sim \frac{\alpha}{R^2} \quad \rightarrow \quad t_{th} \sim \frac{R^2}{\alpha}$$

Physical interpretation: The time for the blob to equilibrate scales with R^2 (quadratic dependence), divided by diffusivity; this also known as the **diffusion time**.

3. Terminal rise speed using Stokes drag

3.1 Identify the driving force: buoyancy

The buoyant force acting on a blob of radius R scales with the density difference between the wax and the surrounding liquid:

$$F_b \sim (\Delta\rho) g R^3$$

The buoyancy-producing density difference is

$$\Delta\rho \sim \rho_{w,c} \beta \Delta T$$

Therefore the buoyant force scaling becomes

$$F_b \sim (\rho_{w,c} \beta \Delta T) g R^3$$

3.2 Identify the resisting force: viscous (Stokes) drag

For creeping flow (very small Reynolds number), the drag on a sphere scales as

$$F_D \sim \mu R U$$

The exact Stokes drag is $6\pi\mu R U$, but constant factors (numerical prefactors) are ignored in scaling analysis.

3.3 Terminal velocity condition

At terminal velocity the net force on the blob is zero, so the buoyant force balances viscous drag:

$$F_b \sim F_D$$

Substituting the scalings gives

$$(\Delta\rho) g R^3 \sim \mu R U$$

Solving for the velocity:

$$U \sim \frac{\Delta\rho g R^2}{\mu}$$

Substituting $\Delta\rho \sim \rho_{w,c}\beta\Delta T$ gives

$$U \sim \frac{\rho_{w,c}\beta\Delta T g R^2}{\mu}$$

This is the terminal speed; with the viscosity term in the denominator, this simply means that as the fluid becomes more viscous, then the terminal speed scales in the opposite direction.

4. Estimate lamp height H

4.1 Relate rise time and cooling time

The blob remains buoyant only while it is hotter than the surrounding liquid. Therefore the blob should lose its buoyancy just as it reaches the top of the lamp. This means the rise time should be comparable to the thermal equilibration time.

$$\frac{H}{U} \sim t_{th}$$

4.2 Substitute the Thermal Equilibrium Time

From Part 2,

$$t_{th} \sim \frac{R^2}{\alpha}$$

Substituting into the rise-time relation gives

$$\frac{H}{U} \sim \frac{R^2}{\alpha}$$

Solving for the lamp height:

$$H \sim U \frac{R^2}{\alpha}$$

Physical interpretation: The height of the lamp is proportional to how quickly the wax blobs rise, to how quickly they cool.

Iteration 2 (Stratified Ambient Model)

- *Stratified ambient temperature varies linearly with height.*

1. Effective driving contrast ΔT_{eff} that sets buoyancy

1.1 Define the temperature mismatch

The ambient temperature varies with height according to

$$T(z) = T_h - \Delta T \frac{z}{H}$$

Define the temperature mismatch between the blob and the local ambient temperature as

$$\Delta(t) \equiv T_b(t) - T(z(t)).$$

1.2 Model thermal relaxation of the blob

The blob temperature relaxes toward the local ambient temperature on the thermal diffusion time scale

$$t_{th} \sim \frac{R^2}{\alpha}$$

A simple relaxation model gives

$$\frac{dT_b}{dt} \sim -\frac{T_b - T(z)}{t_{th}}$$

Since $T_b - T(z) = \Delta$, this becomes

$$\frac{dT_b}{dt} \sim -\frac{\Delta}{t_{th}}$$

1.3 Differentiate the mismatch

Differentiate the mismatch definition

$$\Delta(t) = T_b(t) - T(z(t))$$

with respect to time:

$$\frac{d\Delta}{dt} = \frac{dT_b}{dt} - \frac{d}{dt}(T(z(t)))$$

Applying the chain rule to the second term gives

$$\frac{d\Delta}{dt} = \frac{dT_b}{dt} - \frac{dT}{dz} \frac{dz}{dt}$$

1.4 Substitute the temperature gradient and rise speed

From the ambient temperature profile,

$$\frac{dT}{dz} = -\frac{\Delta T}{H}$$

The blob rises with speed U , so

$$\frac{dz}{dt} = U$$

Substituting these relations and the relaxation model into the mismatch equation gives

$$\frac{d\Delta}{dt} \sim -\frac{\Delta}{t_{th}} + \frac{\Delta T}{H} U$$

1.5 Estimate the effective driving contrast

For a typical steady mismatch, the two terms balance:

$$0 \sim -\frac{\Delta}{t_{th}} + \frac{\Delta T}{H} U$$

Solving for the mismatch gives the **effective driving temperature difference**

$$\Delta T_{eff} \sim \Delta T \frac{U t_{th}}{H}$$

2. Restate the scalings for R , t_{th} , U , and H using the effective driving contrast ΔT_{eff}

From Iteration 2, Part 1 the effective driving temperature mismatch is

$$\Delta T_{eff} \sim \Delta T \frac{U t_{th}}{H}$$

The buoyancy-producing density difference therefore scales as

$$\Delta \rho \sim \rho_{w,c} \beta \Delta T_{eff}$$

Blob radius R (buoyancy vs surface tension)

Balance buoyancy and capillarity:

$$(\Delta \rho) g R^3 \sim \sigma R$$

This gives

$$R \sim \left(\frac{\sigma}{\Delta \rho g} \right)^{1/2} \sim \left(\frac{\sigma}{\rho_{w,c} \beta \Delta T_{eff} g} \right)^{1/2}$$

Thermal equilibration time t_{th} (diffusion across the blob)

$$t_{th} \sim \frac{R^2}{\alpha}$$

Terminal rise speed U (Stokes drag balance)

Balance net buoyancy and viscous drag:

$$(\Delta \rho) g R^3 \sim \mu R U$$

This gives

$$U \sim \frac{\Delta \rho g R^2}{\mu} \sim \frac{\rho_{w,c} \beta \Delta T_{eff} g R^2}{\mu}$$

Lamp height H (rise time comparable to cooling time)

Require the blob to lose buoyancy near the top:

$$\frac{H}{U} \sim t_{th}$$

Thus

$$H \sim U t_{th} \sim \left(\frac{\Delta \rho g R^2}{\mu} \right) \left(\frac{R^2}{\alpha} \right) \sim \frac{\Delta \rho g R^4}{\mu \alpha} \sim \frac{\rho_{w,c} \beta \Delta T_{eff} g R^4}{\mu \alpha}$$

Problem 2. Pressure, Density, Incompressibility, and the Speed of Sound

Derive an evolution equation for the pressure in a fluid starting from the **mass conservation equation**:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \quad (2-1)$$

Analysis

1. Expand the divergence term and use the definition of the speed of sound c in a fluid

$$c^2 = \left(\frac{\partial p}{\partial \rho} \right)_s \quad (2-2)$$

- The subscript s indicates constant entropy, or an isentropic fluid system ("no **heat** is transferred to/from the system to the surroundings").

$$\nabla \cdot (\rho \vec{u}) = \rho (\nabla \cdot \vec{u}) + \vec{u} \cdot (\nabla \rho) \quad (2-3)$$

First principles: Density is a function of pressure and temperature (and **entropy** change is dependent on temperature difference).

$$\rightarrow \rho = \rho(p, T) \quad \rightarrow \rho = \rho(p, s) \quad (2-4)$$

The **full differential** of (2-4):

$$d\rho = \left(\frac{\partial \rho}{\partial p} \right)_s dp + \left(\frac{\partial \rho}{\partial s} \right)_p ds \quad (2-5)$$

$$\text{Assuming constant entropy: } ds = 0 \rightarrow d\rho = \left(\frac{\partial \rho}{\partial p} \right)_s dp \quad (2-6)$$

and *inverting* (2-2) and plugging into (2-6):

$$d\rho = \frac{1}{c^2} dp \quad \rightarrow \quad \nabla \rho = \frac{1}{c^2} \nabla p \quad (2-7)$$

Physical interpretation: Density changes is proportional to pressure changes through the speed of sound (squared), c^2 .

2. Convert this into a relation between material derivatives.

Applying the chain rule:

$$\frac{d\rho}{dt} = \frac{d\rho}{dp} \cdot \frac{dp}{dt} \quad \rightarrow \quad \frac{D\rho}{Dt} = \frac{d\rho}{dp} \cdot \frac{Dp}{Dt} \quad \rightarrow \quad \frac{D\rho}{Dt} = \frac{1}{c^2} \cdot \frac{Dp}{Dt} \quad (2-8)$$

"What does the **isentropic assumption** tell us?"

The thermodynamic relation is isentropic, so density changes relate to pressure changes through c^2 , and we can use apply that relationship along the particle path.

3. For incompressible flow, what does mass conservation tell us about the dynamics of p ?

Compressibility & The Speed of Sound

Since $\left(\frac{\partial \rho}{\partial p}\right)_s = \frac{1}{c^2}$, this is a **measure of compressibility**, which indicates how much density changes when pressure changes.

In incompressible flow, $\frac{\partial \rho}{\partial p} = 0$

then, $\frac{1}{c^2} = 0 \quad \leftarrow \quad \text{only possible as } c \rightarrow \infty$

From (2-1) and (2-3):

$$\frac{\partial \rho}{\partial t} + \vec{u} \cdot \nabla \rho + \rho \nabla \cdot \vec{u} = 0 \quad \rightarrow \quad \frac{D\rho}{Dt} + \rho \nabla \cdot \vec{u} = 0 \quad (2-9)$$

Plug (2-8) into (2-9):

$$\frac{Dp}{Dt} = -c^2 \rho \nabla \cdot \vec{u} \quad (2-10)$$

and for incompressible flow: $\nabla \cdot \vec{u} = 0$

(2-10) then becomes $\rightarrow \frac{Dp}{Dt} = 0 \quad (2-11)$

Physical interpretation:

Noting the following fundamental principles --

Continuity Equation governs mass conservation → gives **velocity scaling**.

Momentum (Navier-Stokes) Equation governs force balance → gives **pressure field**.

Mass conservation, with the incompressible flow assumption, does not determine p .

In incompressible flow, continuity becomes a **constraint on the velocity field**, and pressure is a parameter that **enforces** that constraint through the momentum equation. So p is set by dynamics/momentum + boundary conditions, not by continuity alone. [1]

Problem 3. Vortex stretching in an axisymmetric stagnation flow

First, let's get a "birds eye view" of this flow field:

```
%% HW4 - Problem 3 Flow Visualization (Axisymmetric straining flow)
% u_r = -(alpha/2) r
% u_z =  alpha z
% No swirl (u_theta = 0)
%
% This script uses the Symbolic Math Toolbox to define the field, then
% plots:
% 1) a 2D quiver plot in the r-z plane (axisymmetric meridional plane)
% 2) streamlines in the r-z plane
% 3) an optional 3D visualization by revolving the r-z field about z-axis

clear; clc; close all;

%% --- Symbolic definitions ---
syms r z alpha real
ur = -(alpha/2)*r;
uz =  alpha*z;

% Convert to numeric function handles for plotting
ur_fun = matlabFunction(ur, "Vars", [r z alpha]);
uz_fun = matlabFunction(uz, "Vars", [r z alpha]);

%% --- Parameters (edit as needed) ---
alpha_val = 1.0;      % [1/s] strain rate (pick a convenient value)
Rmax = 2.0;           % max r extent
Zmax = 2.0;           % max |z| extent

Nr = 25;              % grid density in r
Nz = 31;              % grid density in z

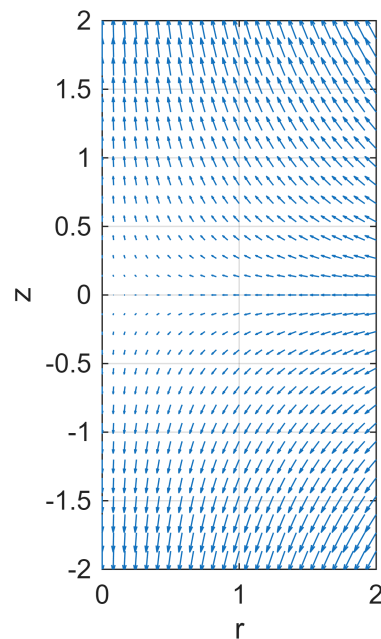
%% --- Grid in meridional (r,z) plane ---
r_vec = linspace(0, Rmax, Nr);      % r >= 0
z_vec = linspace(-Zmax, Zmax, Nz);
[R, Z] = meshgrid(r_vec, z_vec);
```

```

UR = ur_fun(R, Z, alpha_val);
UZ = uz_fun(R, Z, alpha_val);

%% --- Plot 1: Quiver plot in r-z plane ---
fig = figure;
fig.Theme = "light";
quiver(R, Z, UR, UZ, 1.2);
grid on; axis equal tight;
xlabel('r'); ylabel('z');
title('Axisymmetric straining flow in meridional plane: (u_r, u_z)');
xlim([0 Rmax]); ylim([-Zmax Zmax]);

```



```

%% --- Plot 2: Streamlines in r-z plane ---
% Create a finer grid for smoother streamlines
r_vec_f = linspace(0, Rmax, 80);
z_vec_f = linspace(-Zmax, Zmax, 120);
[Rf, Zf] = meshgrid(r_vec_f, z_vec_f);

URf = ur_fun(Rf, Zf, alpha_val);
UZf = uz_fun(Rf, Zf, alpha_val);

% Seed points for streamlines (pick a set across the domain)
start_r = linspace(0.1, Rmax, 10);
start_z = -Zmax*ones(size(start_r)); % start along bottom edge
startPts = [start_r(:), start_z(:)];

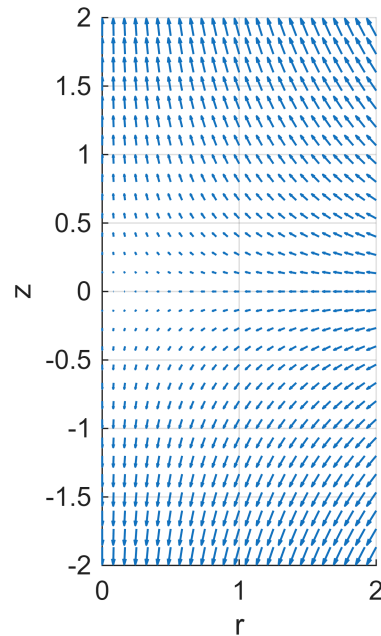
fig = figure;
fig.Theme = "light";
hold on;

```

```

quiver(R, Z, UR, UZ, 1.0, 'LineWidth', 0.75); % context arrows
streamline(stream2(Rf, Zf, URf, UZf, startPts(:,1), startPts(:,2)));
grid on; axis equal tight;
xlabel('r'); ylabel('z');
title('Streamlines in meridional plane (r-z)');
xlim([0 Rmax]); ylim([-Zmax Zmax]);
hold off;

```



```

%% --- Plot 3 (Optional): 3D "revolved" view about z-axis ---
% We revolve the meridional plane around the z-axis:
% r = sqrt(x^2 + y^2), and the velocity components become:
% u_x = u_r * (x/r), u_y = u_r * (y/r), u_z = u_z
do3D = true;

if do3D
    % 3D grid (moderate size to keep it readable)
    x_vec = linspace(-Rmax, Rmax, 13);
    y_vec = linspace(-Rmax, Rmax, 13);
    z_vec3 = linspace(-Zmax, Zmax, 11);
    [X, Y, Z3] = meshgrid(x_vec, y_vec, z_vec3);

    R3 = sqrt(X.^2 + Y.^2);

    % Avoid division by zero on the axis (r=0): set direction factors to 0
    ex = zeros(size(R3)); ey = zeros(size(R3));
    mask = R3 > 1e-12;
    ex(mask) = X(mask)./R3(mask);
    ey(mask) = Y(mask)./R3(mask);

    UR3 = ur_fun(R3, Z3, alpha_val);

```



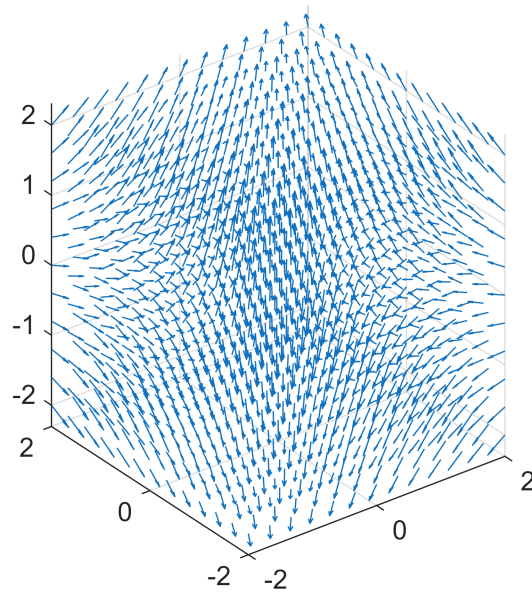
```

UZ3 = uz_fun(R3, Z3, alpha_val);

UX = UR3 .* ex;
UY = UR3 .* ey;

fig = figure;
fig.Theme = "light";
quiver3(X, Y, Z3, UX, UY, UZ3, 1.2);
grid on; axis equal tight;
xlabel('x'); ylabel('y'); zlabel('z');
title('3D visualization of axisymmetric straining flow (revolved about z-
axis)');
end

```



1. Evolution Equation for Azimuthal Vorticity ω_θ

1.1 Start from the incompressible vorticity equation

For an incompressible fluid with constant kinematic viscosity ν ,

$$\frac{\partial \vec{\omega}}{\partial t} + (\vec{u} \cdot \nabla) \vec{\omega} = (\vec{\omega} \cdot \nabla) \vec{u} + \nu \nabla^2 \vec{\omega}$$

where the vorticity vector is defined as

$$\vec{\omega} = \nabla \times \vec{u} \quad (\text{Ref: [1, p. 144]})$$

1.2 Specialize to axisymmetric flow with no swirl

For axisymmetric flow,

$$\frac{\partial}{\partial \theta} = 0$$

and "no swirl" means

$$u_\theta = 0$$

In this situation the relevant vorticity component is the azimuthal component

$$\omega_\theta(r, z, t)$$

The material derivative acting on ω_θ becomes

$$\frac{D\omega_\theta}{Dt} = \frac{\partial \omega_\theta}{\partial t} + u_r \frac{\partial \omega_\theta}{\partial r} + u_z \frac{\partial \omega_\theta}{\partial z}$$

1.3 Identify the stretching term

For axisymmetric flow without swirl, the θ component of the vorticity stretching term simplifies to

$$[(\vec{\omega} \cdot \nabla)\mathbf{u}]_\theta = \frac{u_r}{r} \omega_\theta$$

This term represents the geometric stretching or compression of vorticity due to radial convergence or divergence of the flow.

1.4 Viscous diffusion term in cylindrical coordinates

The Laplacian acting on the azimuthal vorticity component in axisymmetric cylindrical coordinates is

$$[\nabla^2 \vec{\omega}]_\theta = \frac{\partial^2 \omega_\theta}{\partial r^2} + \frac{1}{r} \frac{\partial \omega_\theta}{\partial r} + \frac{\partial^2 \omega_\theta}{\partial z^2} - \frac{\omega_\theta}{r^2}$$

Final evolution equation for ω_θ

Combining convection, stretching, and viscous diffusion gives

$$\frac{\partial \omega_\theta}{\partial t} + u_r \frac{\partial \omega_\theta}{\partial r} + u_z \frac{\partial \omega_\theta}{\partial z} = \frac{u_r}{r} \omega_\theta + \nu \left(\frac{\partial^2 \omega_\theta}{\partial r^2} + \frac{1}{r} \frac{\partial \omega_\theta}{\partial r} + \frac{\partial^2 \omega_\theta}{\partial z^2} - \frac{\omega_\theta}{r^2} \right)$$

Substitute the given straining flow

The velocity field is

$$u_r = -\frac{\alpha}{2} r, \quad u_z = \alpha z$$

Substituting into the vorticity equation gives

$$\frac{\partial \omega_\theta}{\partial t} - \frac{\alpha}{2} r \frac{\partial \omega_\theta}{\partial r} + \alpha z \frac{\partial \omega_\theta}{\partial z} = -\frac{\alpha}{2} \omega_\theta + \nu \left(\frac{\partial^2 \omega_\theta}{\partial r^2} + \frac{1}{r} \frac{\partial \omega_\theta}{\partial r} + \frac{\partial^2 \omega_\theta}{\partial z^2} - \frac{\omega_\theta}{r^2} \right)$$

2. Solve the inviscid ω_θ equation

2.1 Start from the inviscid vorticity equation

Neglect viscosity ($\nu = 0$). The azimuthal vorticity equation from Part 1 becomes

$$\frac{\partial \omega_\theta}{\partial t} + u_r \frac{\partial \omega_\theta}{\partial r} + u_z \frac{\partial \omega_\theta}{\partial z} = \frac{u_r}{r} \omega_\theta$$

For the given straining flow

$$u_r = -\frac{\alpha}{2} r, \quad u_z = \alpha z,$$

so that

$$\frac{u_r}{r} = -\frac{\alpha}{2}$$

Substituting these expressions gives

$$\frac{\partial \omega_\theta}{\partial t} - \frac{\alpha}{2} r \frac{\partial \omega_\theta}{\partial r} + \alpha z \frac{\partial \omega_\theta}{\partial z} = -\frac{\alpha}{2} \omega_\theta$$

2.2 Follow a fluid particle (method of characteristics)

Along a particle trajectory $(r(t), z(t))$, the vorticity evolves according to the material derivative

$$\frac{D\omega_\theta}{Dt} = -\frac{\alpha}{2} \omega_\theta$$

The particle motion itself satisfies

$$\frac{dr}{dt} = u_r = -\frac{\alpha}{2} r, \quad \frac{dz}{dt} = u_z = \alpha z$$

Solving these equations gives the particle paths

$$r(t) = r_0 e^{-\alpha t/2}, \quad z(t) = z_0 e^{\alpha t}$$

The initial particle labels can therefore be written as

$$r_0 = r e^{\alpha t/2}, \quad z_0 = z e^{-\alpha t}$$

2.3 Solve the vorticity evolution

The vorticity along a particle path satisfies

$$\frac{D\omega_\theta}{Dt} = -\frac{\alpha}{2} \omega_\theta$$

Integrating gives

$$\omega_\theta(t) = \omega_\theta(0) e^{-\alpha t/2}$$

The initial value is the initial vorticity field evaluated at the particle's starting position:

$$\omega_{\theta}(0) = \omega_{\theta,0}(r_0, z_0)$$

Therefore

$$\omega_{\theta}(r(t), z(t), t) = e^{-\alpha t/2} \omega_{\theta,0}(r_0, z_0)$$

Substituting the expressions for r_0 and z_0 **yields**

$$\omega_{\theta}(r, z, t) = e^{-\alpha t/2} \omega_{\theta,0}(r e^{\alpha t/2}, z e^{-\alpha t})$$

```
%% Visualization of vorticity evolution in axisymmetric straining flow
% wtheta(r,z,t) = exp(-alpha t/2) * wtheta( r*exp(alpha t/2), z*exp(-alpha t) )

clear; clc; close all;

%% --- Symbolic definition ---
syms r z t alpha real

% Choose an initial vorticity field wtheta0(r,z)
% Example: Gaussian vortex ring
sigma = 0.6;

omega0 = exp(-(r^2 + z^2)/sigma^2);

% Full solution
omega = exp(-alpha*t/2) * subs(omega0, ...
    [r, z], ...
    [r*exp(alpha*t/2), z*exp(-alpha*t)]);

% Convert to numerical function
omega_fun = matlabFunction(omega, "Vars", [r z t alpha]);

%% --- Parameters ---
alpha_val = 1.0;

rmax = 2;
zmax = 2;

Nr = 120;
Nz = 120;

r_vec = linspace(0, rmax, Nr);
z_vec = linspace(-zmax, zmax, Nz);

[R, Z] = meshgrid(r_vec, z_vec);
```

```

%% --- Time loop for visualization ---
figure

for time = linspace(0,2,40)

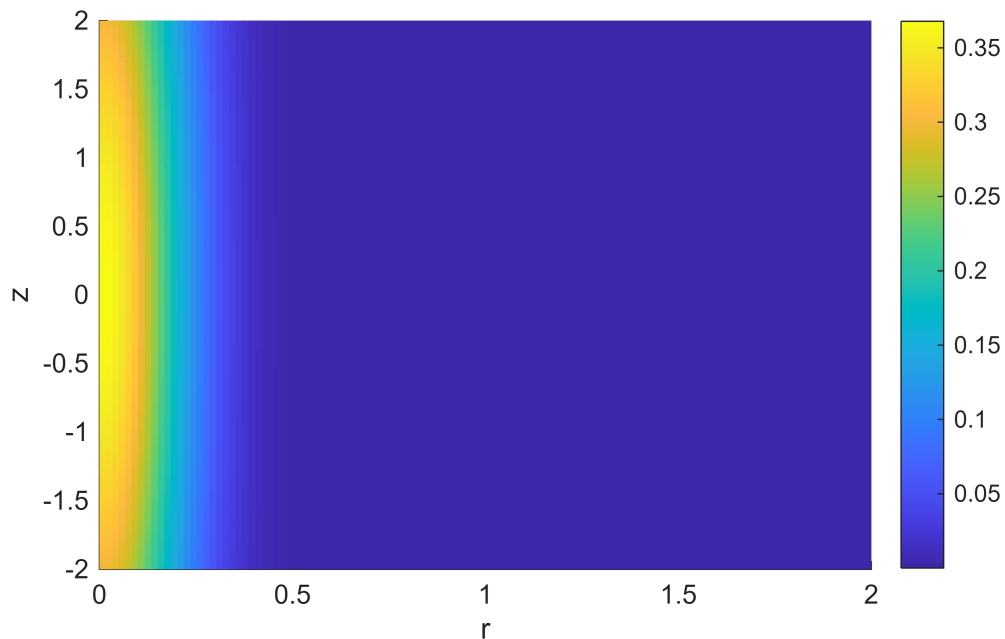
    OMEGA = omega_fun(R,Z,time,alpha_val);

    surf(R,Z,OMEGA,'EdgeColor','none')
    view(2)
    colorbar
    axis tight
    xlabel('r')
    ylabel('z')
    title(['\omega_\theta(r,z,t),  t = ',num2str(time)])

    drawnow

end

```



References

- [1] Cengel, Y. (2018). *Fluid Mechanics: Fundamentals and Applications* (4th ed.). New York: McGraw-Hill.
- [2] Kreyszig, E. (1999). *Advanced Engineering Mathematics* (8th ed.). New York: John Wiley & Sons.